

Data Analytics Capstone

**AI-Driven Modelling of Sustainable Development Goal Indicators: A South
Asia-Focused Study Using Open-Access UN SDG Data**

Synopsis

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Format of the Synopsis

Introduction

The Sustainable Development Goals (SDGs) remain the world's roadmap for ending poverty, protecting the planet and tackling inequalities. In the year 2015 leaders of 193 countries under United Nations made a historic promise to secure the rights and well-being of everyone on a healthy, thriving planet when they adopted the 2030 Agenda for Sustainable Development and its 17 Sustainable Development Goals (SDGs). 17 SDGs offer the most practical and effective pathway to tackle the causes of violent conflict, inequalities, climate change and environmental degradation and aim to ensure that no one will be left behind. They reflect an understanding that sustainable development everywhere must integrate economic growth, social well-being and environmental protection.(UNO, 2015)

Sustainable development is one of the most pressing challenges of the 21st century, underscored by global initiatives such as the 2030 Agenda of the United Nations, which outlines 17 SDGs aimed at tackling critical issues such as poverty, inequality, environmental degradation, and climate change

Figure 1**UN Sustainable Development Goals**

Each of the 17 goals have Targets and each target has measurable indicators which are defined by United Nations SDG goal and for these 17 goals there are 234 unique indicators as part of the Global indicator framework for the Sustainable Development Goals and targets of the 2030 Agenda for Sustainable Development(UNO, 2020) . The Metadata for this are defined in UN SDG portal UN SDG Indicators Metadata Repository (UNO, 2026b)

Indicators for the 17 SDGs pool together a wide range of issues—hunger, poverty, inequality, health, species facing extinction, land degradation, gender inequality, gaps in education quality,

productivity and technological achievements. These issues span across sectors and regions and our sustainability requires an adaptive understanding of their triggers. It is in that context that we view them as highly voluminous, volatile and dynamic data attributes, the behaviour and variations of which we need to track and understand in a unified and interdisciplinary manner.(Mwitondi et al., 2020)

SDG Indicators data is Voluminous and dynamic requiring a SDG Modelling and Data Visualization with Predictive power through AI Integration for the UN , Policy Makers and Government Leaders to understand the impact of their implementation programs. The data for the different indicators are available in the UN SDG Global Datahub which is open access(UNO, 2026a) and the data is available for different indicators from the years 2015 to 2026.

. This study focuses on SDG 9 (Industry, Innovation, and Infrastructure) and SDG 11 (Sustainable Cities and Communities), which are pivotal for the socio-economic advancement of South Asia. The region, encompassing India, Pakistan, Bangladesh, Nepal, Sri Lanka, Bhutan, and Maldives, is characterized by rapid urbanization, industrial growth, and significant infrastructure challenges. Leveraging open-access datasets from the United Nations Statistics Division (UNSD), this research employs advanced AI-driven modelling techniques to analyze SDG indicators, forecast future trends, and uncover critical factors influencing sustainable development outcomes.

South Asia faces multifaceted challenges including infrastructure deficits, uneven industrialization, urban congestion, environmental pollution, and heightened vulnerability to climate change impacts. By integrating machine learning models such as Random Forest,

Gradient Boosting, and Long Short-Term Memory (LSTM) networks with traditional statistical methods, this study aims to provide robust predictive insights and actionable recommendations for policymakers.

This synopsis presents a comprehensive overview of the study's background, problem statement, research questions, methodology, sample size determination, analytical strategies, and anticipated contributions to both academic research and policy formulation

Background and Context

South Asia is home to nearly one-fourth of the world's population and is undergoing rapid economic transformation. Industrialization and urbanization are accelerating, but infrastructure development and sustainable city planning lag behind. SDG 9 emphasizes resilient infrastructure, sustainable industrialization, and innovation capacity. SDG 11 focuses on inclusive, safe, resilient, and sustainable cities. Together, these goals shape the region's long-term development trajectory.

Despite progress, South Asian countries face persistent challenges: inadequate transport networks, limited digital infrastructure, rising urban slums, air pollution, waste management issues, and vulnerability to climate-induced disasters. The UN SDG database provides annual country-level indicators such as manufacturing value added, research and development expenditure, urban population living in slums, access to public transport, and waste collection efficiency. These indicators allow researchers to analyze disparities, identify bottlenecks, and evaluate policy effectiveness.

AI-driven modelling can uncover hidden patterns, forecast SDG progress, and support evidence-based decision-making. Machine learning models such as Random Forest, Gradient Boosting, and LSTM time-series forecasting can complement traditional statistical methods like

regression, ANOVA, and correlation analysis. This hybrid approach enhances predictive accuracy and interpretability.

Problem Statement

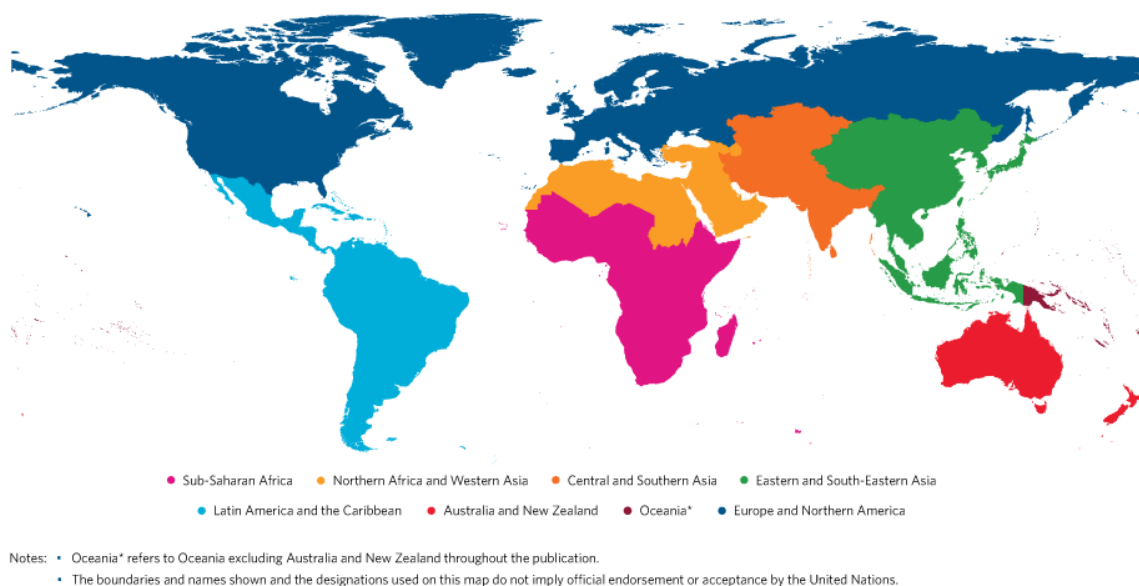
The complexity of SDGs interactions and dynamics renders itself readily to the problem of data randomness, identifying triggers of the indicators amounts to uncovering *what works* in different sectors and countries which, given the spatio-temporal variations can be challenging.

(Mwitondi et al., 2020)

South Asian countries exhibit uneven progress toward SDG 9 and SDG 11, with significant gaps in infrastructure quality, industrial innovation, and sustainable urban development. Existing analyses are descriptive and lack predictive modelling. Policymakers need data-driven insights to understand key drivers, forecast future SDG performance, and design targeted interventions. This study addresses the gap by applying hybrid statistical and AI modelling techniques to open-access SDG data for South Asia.

Purpose of the Study

As per UN the Globe is divided into following regional groupings for data collection and tracing purpose

Figure 2**Regional groupings used for SDG Indicators Reporting**

Source: <https://unstats.un.org/sdgs/indicators/regional-groups/>

This work looks at variations in SDG data across sectors from the South Asian Countries comprising of India, Bhutan, Sri Lanka, Bangladesh, Afghanistan, Iran, Nepal, Pakistan and Maldives.

The purpose of this study is to develop predictive, explanatory, and comparative models for SDG 9 and SDG 11 indicators across South Asian countries using open-access UN SDG datasets. The study aims to identify key drivers, forecast future trends, and compare

country-level and regional-level performance using hybrid statistical and machine learning approaches.

Scope and objectives

To uncover triggers of the indicators, a general pragmatic approach to examine similarities and dissimilarities among data attributes is adapted that could lead to uncovering potentially useful information in the attributes. The following are the Research Objectives of this project

1. To exhibit the impact of data randomness and variations through data visualisation.
2. To promote interdisciplinary activities for problem identification and attainment of agenda 2030.
3. To highlight and support interdisciplinary paths for a unified understanding of global phenomena.

RQ1: What are the key statistical relationships between SDG 9 and SDG 11 indicators across South Asian countries?

RQ2: Can machine learning models accurately predict SDG 9 and SDG 11 indicator values for South Asian countries?

RQ3: What are the most influential drivers affecting SDG 9 and SDG 11 performance in the region?

RQ4: How do country-level SDG trajectories compare with regional-level aggregated trends?

Sample size calculation

<Will be completed in the Final Version>

Data Description

Data Sources (Open Access)

- UN SDG Global Database (<https://unstats.un.org/sdgs/dataportal/database>)
- UN Data Portal (<https://data.un.org>)
- UN Urban Indicators (<https://urban-data-guo-un-habitat.hub.arcgis.com/>)
- UN Habitat Data <https://data.unhabitat.org/>
- UN Industrial Development Indicators (<https://stat.unido.org>)

Data dictionary (mandatory)

Preliminary data is loaded in the github and can be accessed from the link

<https://github.com/srinivastnptel-collab/Data-Analytics-Capstone/upload/main/data/raw>

Table 1

Core Columns in the UN SDG Dataset

Column Name	Meaning
Goal	SDG number (e.g., 9 = Industry, Innovation & Infrastructure; 11 = Sustainable Cities & Communities).
Target	Specific SDG target under the goal (e.g., 9.2 = Promote inclusive industrialization).
Indicator	Official SDG indicator code (e.g., 9.2.1).
SeriesCode	Internal UN code for the indicator series.
SeriesDescription	Full description of the indicator (e.g., Manufacturing value added as % of GDP).
GeoAreaCode	Numeric country code (UN standard).
GeoAreaName	Country name (India, Nepal, Sri Lanka, etc.).
TimePeriod	Year of observation (2009–2021 in your dataset).
Value	Numeric value of the indicator for that year and country.
Units	Measurement units (% , USD, per capita, etc.).
Footnotes	Notes on data quality, missing values, or estimation.
Source	UN agency responsible for the indicator (UNIDO, UN Habitat, World Bank, etc.).

These columns appear in every SDG dataset, regardless of the goal.

Table 2**SDG 9 — Industry, Innovation & Infrastructure**

Indicator Code	Indicator Name	Description	Unit
9.1.1	Rural access index	Proportion of rural population living within 2 km of an all-season road	%
9.1.2	Passenger & freight volumes	Transport volumes by mode	Ton-km / Passenger-km
9.2.1	Manufacturing value added	MVA as % of GDP	%
9.2.2	Manufacturing employment	Manufacturing employment as % of total employment	%
9.3.1	MSME share in GDP	Contribution of small industries to GDP	%
9.3.2	MSME share in employment	Employment in MSMEs as % of total	%
9.4.1	CO ₂ emissions per unit of GDP	Environmental efficiency of industry	kg CO ₂ per USD
9.5.1	R&D expenditure	Research expenditure as % of GDP	%
9.5.2	Researchers per million	Number of researchers per million population	Count
9.a.1	Infrastructure support	International financial support for infrastructure	USD
9.b.1	Medium/high-tech manufacturing share	Share of high-tech industries in total manufacturing	%
9.c.1	Mobile network coverage	Population covered by mobile networks	%

Table 3

SDG 11 — Sustainable Cities & Communities

Indicator Code	Indicator Name	Description	Unit
11.1.1	Urban slum population	Proportion of urban population living in slums	%
11.2.1	Access to public transport	Population with convenient access to public transport	%
11.3.1	Land consumption vs population growth	Urban expansion efficiency	Ratio
11.3.2	Urban planning participation	Proportion of cities with participatory planning	%
11.5.1	Disaster mortality	Number of deaths due to disasters	Count
11.5.2	Economic loss due to disasters	Economic loss as % of GDP	%
11.6.1	Waste collection coverage	Solid waste collection efficiency	%
11.6.2	Air quality (PM2.5)	Annual mean PM2.5 concentration	µg/m ³
11.7.1	Public open space	Share of built-up area that is open public space	%
11.7.2	Safety perception	Proportion of population feeling safe walking alone	%
11.a.1	Urban-rural linkage policies	National urban policies	Yes/No
11.b.1	Disaster risk reduction strategies	Local DRR strategies	Count
11.b.2	DRR implementation	Proportion of local governments implementing DRR	%
11.c.1	Sustainable building support	Financial support for sustainable buildings	USD

Analytic approach

The article (Greif et al., 2025) provides details of the AI Techniques used in the SDG Modelling .This document will be referred during the Modelling process. The latest SDG Report provides information on AI usage in the SDG data generation and will also be used as guideline(UNO, 2026c)

Techniques Used

- Descriptive statistics
- Correlation analysis
- Multiple regression
- ANOVA
- Random Forest
- Gradient Boosting
- LSTM time-series forecasting
- Feature importance analysis

Hypotheses

- H1: SDG 9 indicators significantly influence SDG 11 outcomes.
- H2: ML models outperform statistical models in predicting SDG indicators.
- H3: Country-level disparities significantly affect regional SDG trends.

Solution to RQ1-4

RQ1: Statistical Techniques

- Correlation matrix
- Regression models

- ANOVA for country comparisons

RQ2: Machine Learning Techniques

- Random Forest
- XGBoost
- LSTM forecasting

RQ3: Driver Analysis

- Feature importance
- SHAP values
- Regression coefficients

RQ4: Comparative Modelling

- Country-level vs regional-level time-series
- Cluster analysis

Evaluation Metrics

- RMSE
- MAE
- R^2
- MAPE
- Confusion matrix (if classification used)

Recommendation and application

Target Users

- Policymakers
- UN agencies

- Urban planners
- Infrastructure ministries
- Development economists

Applications

- Forecasting SDG progress
- Identifying priority intervention areas
- Designing targeted infrastructure and urban policies
- Benchmarking country performance

References and bibliography

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